



FACULTY OF ENGINEERING AND BASIC SCIENCES

Project - Extracción, Transformación y carga

ACADEMIC PROGRAM: DATA ENGINEERING AND ARTIFICIAL INTELLIGENCE

Team Members:

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Project Context

This project analyzes the indicator “**Proportion of population with primary reliance on clean fuels and technologies**”, which represents a global issue affecting multiple populations worldwide between 2000 and 2023.

According to several studies conducted by the World Health Organization (WHO), exposure to indoor air pollutants caused by the use of highly contaminating fuels and technologies leads to a wide range of adverse health effects. These include eye diseases, respiratory illnesses, and cancer, particularly in rural areas and developing countries where access to clean energy remains limited.



Data Source Evaluation

Why was the dataset selected?

The dataset was selected because it is a reliable and relevant source for analyzing Sustainable Development Goal 7 (SDG 7), specifically indicator 7.1.2. The data comes from the World Health Organization (WHO), ensuring that it is official, real, and supported by an authoritative international institution.

Additionally, the dataset is available in CSV format, which facilitates its integration into an ETL process and migration into a relational database. Its structured variables and indicators allow for comparative analysis and easy modeling within a Data Warehouse. Furthermore, the dataset has strong potential for future expansion, as it can be integrated with other datasets related to health, energy, or sustainable development.



General Objective

The purpose of this project is to develop a basic ETL process to extract, load, analyze, transform, and visualize a dataset related to SDG 7 (Indicator 7.1.2), using a relational database as the primary source of analysis. The goal is to identify and study patterns or trends in access to clean fuels and technologies for cooking.

Specific Objectives

1. Select a dataset aligned with SDG 7 and explain its relevance to the problem.
2. Load the CSV file into a relational database as a raw dataset, preserving its original structure.
3. Perform Exploratory Data Analysis (EDA) directly from the database, identifying relevant patterns.
4. Apply cleaning and transformation processes (renaming columns, correcting data types, filtering variables, and standardizing values) to generate a transformed dataset.
5. Store the transformed dataset in the database for analysis and visualization.

6. Build a dashboard based on the transformed dataset to answer key project questions.
7. Document the ETL architecture, data model, and results obtained in a technical report hosted on GitHub.

Project Requirements

Functional Requirements

1. Import a CSV file containing the selected dataset (minimum 10,000 rows and 10 columns).
2. Migrate the CSV file to a relational database.
3. Store a raw version of the dataset in the database.
4. Execute queries from the database to perform EDA.
5. Apply transformations to generate a clean and analytical dataset.
6. Store the transformed dataset in the database.
7. Generate charts/dashboard using only data queried from the database.
8. Export results (charts, tables, conclusions) to include them in the final report.
9. Maintain the project versioned in GitHub with a clear structure and informative commits.

Non-Functional Requirements

1. **Reproducibility:** The ETL process must be executable multiple times with consistent results.
2. **Data Integrity:** The migration must preserve records without losses or unjustified modifications.

3. **Project Organization:** Clear folder and file structure (scripts, notebooks, SQL, documentation, etc.).
4. **Code Readability:** Understandable code with appropriate naming conventions and comments when necessary.
5. **Documentation:** Inclusion of a README file, justification of the technology stack, and execution instructions.
6. **Version Control:** Proper use of Git, GitHub, and a correctly configured .gitignore file.
7. **Traceability:** Clear differentiation between the raw dataset and the transformed dataset.
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Dataset Requirements

The selected dataset must meet the following criteria:

1. Be aligned with at least one SDG (SDG 7).
2. Contain a minimum of 10,000 rows.
3. Contain at least 10 columns.
4. Be available in a processable format, preferably CSV.
5. Include variables useful for temporal, geographical, and/or categorical analysis.
6. Allow justification of its availability, relevance, structure, and quality.

Source:

<https://www.who.int/data/gho/data/themes/air-pollution/household-air-pollution>

Analysis Requirements (EDA and Visualization)

The analysis must include, at minimum:

1. Review of null values.
2. Review of data types.
3. General description of variables.
4. Identification of relevant trends and differences.
5. Visualizations that support the project objectives.

Note: All data queries must be executed directly from the database, not from the CSV file



Architecture Requirements (Based on the Proposed Process)

The project architecture will follow this workflow:

1. CSV → Raw Dataset (Database)
2. EDA from the database
3. Transformation and cleaning → Transformed Dataset (Database)
4. Visualization / Dashboard (from the transformed dataset)

Scope of This First Delivery



In this first delivery, the following will be developed:

1. Selection and justification of the dataset in relation to the SDG.
2. Evaluation of the data source.

3. Basic ETL architecture design.
4. Migration of the CSV file to a relational database.
5. Initial EDA.
6. Main transformations.
7. Visualizations / dashboard.

GITHUB repository

<https://github.com/juandavidmeza-prog/etl-project-first-delivery>